

# How Markets Absorb News

**A cross-announcement study of price discovery mechanisms following scheduled and unscheduled information releases**

*Project Proposal*

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## 1. Abstract

We are interested in how prices absorb news at the microstructural level. Two mechanisms are possible: market makers revise quotes the instant an announcement arrives, or the market learns gradually from the order flow of traders acting on their own interpretations. In high-frequency data the two are separable: a quote gap before any trade versus cumulative return explained by signed flow. We want to measure, for each announcement, the fraction of the price adjustment attributable to quote revision, and to test whether this fraction falls as news gets harder to interpret, highest for a single-number release against a published consensus and lowest for extemporaneous speech, using analyst forecast dispersion as the continuous proxy. Identification would come from variation across announcement types: scheduled macro releases, central bank communication, earnings, and unscheduled news, observed in liquid futures markets with sub-second timestamps.

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## 2. Motivation

Most of the empirical literature on announcement effects treats the release as atomic: measure the price a few minutes before, measure it a few minutes after, and study the difference. That approach, standard since Balduzzi, Elton and Green (2001) and refined in the monetary policy setting by Kuttner (2001) and Gürkaynak, Sack and Swanson (2005), is appropriate for identifying the *magnitude* of a shock's effect. It deliberately discards the interior of the event window.

The interior is where the mechanism lives. Two announcements can produce identical five-minute price changes while getting there completely differently: one through a single instantaneous quote gap with almost no trading, the other through several minutes of heavy, informative order flow. Theory says these correspond to different informational environments. Nobody has systematically mapped which announcements fall where.

The obstacle has been that within a single announcement type there is little variation in the informational environment: every CPI release is a CPI release. The solution is to widen the sample across announcement types and treat announcement characteristics as covariates. This turns a descriptive microstructure exercise into a testable cross-sectional theory.

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### 3. The Announcement Taxonomy

Announcements are classified along four dimensions, each of which generates a prediction about the discovery mechanism.

**Dimension 1: Scheduling.** Is the arrival time known in advance?

- *Scheduled to the second*: macro data releases, central bank decisions, scheduled earnings.
- *Scheduled to the day or window*: central bank speeches, some regulatory decisions.
- *Unscheduled*: geopolitical developments, corporate incidents, unexpected policy action.

Scheduling determines whether liquidity providers can withdraw in advance. Under scheduled releases we expect a characteristic pre-announcement depth collapse and a wide, thin book at the moment of arrival; under unscheduled news, no pre-event withdrawal but a sharper post-event liquidity shock.

**Dimension 2: Representation.** Is the news a number or a narrative?

- *Scalar numeric*: a headline inflation print, a payrolls figure, a policy rate decision.
- *Multi-dimensional numeric*: a payrolls report with revisions, participation and wage components; an earnings release with revenue, margin and guidance.
- *Narrative text*: a policy statement, a regulatory ruling, a court decision.
- *Extemporaneous speech*: a press conference, an earnings call Q&A.

This is the primary predictor. Interpretation heterogeneity rises monotonically down this list, and with it the expected share of the price move that flows through order flow rather than quote revision.

**Dimension 3: Ex-ante disagreement.** For releases with a published forecast survey, the cross-sectional dispersion of analyst forecasts is an observable, continuous proxy for how much participants are likely to disagree *after* the release as well as before. This is the cleanest available instrument for interpretation heterogeneity and requires no judgment call from the researcher.

**Dimension 4: Surprise magnitude.** The standardized deviation of the realized value from consensus, which enters as a control and interacts with the mechanism split.

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## 4. Research Questions

**Q1, Mechanism (primary).** What fraction of the total price adjustment occurs in the initial quote revision, before any trade takes place, versus accumulating through subsequent signed order flow? How does this fraction vary across the announcement taxonomy?

**Q2, Speed.** How long until the efficient price ceases to move? Does the discovery horizon scale with surprise magnitude, with ex-ante disagreement, or with both?

**Q3, Location.** Which instrument leads the adjustment, and does the leader change with announcement type? Rate-sensitive releases should be led by rates instruments; broad risk news by index futures.

**Q4, Liquidity dynamics.** How does depth at touch evolve around scheduled versus unscheduled arrivals, and does the pre-event withdrawal pattern predict the post-event discovery mechanism?

**Q5, Asymmetry.** Do good-news and bad-news surprises of equal standardized magnitude discover at the same speed?

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## 5. Hypotheses

| #  | Hypothesis                                                        | Predicted result                                               |
|----|-------------------------------------------------------------------|----------------------------------------------------------------|
| H1 | Scalar numeric releases are impounded by quote revision           | Pre-trade quote jump exceeds 70% of the five-minute move       |
| H2 | Narrative and extemporaneous releases are impounded by order flow | Pre-trade quote jump below 40%; remainder loads on signed flow |
| H3 | The order-flow share rises in ex-ante forecast dispersion         | Positive coefficient, holding surprise magnitude fixed         |

|    |                                                                        |                                                                            |
|----|------------------------------------------------------------------------|----------------------------------------------------------------------------|
| H4 | Scheduled arrivals show pre-event depth withdrawal; unscheduled do not | Divergent depth paths in the 60 s before arrival                           |
| H5 | Discovery horizon increases in surprise magnitude                      | Positive coefficient of convergence time on absolute standardized surprise |
| H6 | Bad news discovers more slowly than good news                          | Asymmetric convergence horizon, conditional on magnitude                   |

Every hypothesis carries a placebo counterpart at the same clock time on non-announcement days. An effect that survives on placebo days is a time-of-day artifact, not a news effect.

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## 6. Sample Design

The sample is built to populate every cell of the taxonomy rather than to maximize observations of any one release.

**Scheduled numeric releases.** The major macro calendar over a five-to-ten year window: employment, inflation, growth, activity and sentiment indicators, spanning both the early-morning and mid-morning release slots. Several hundred to a few thousand events depending on how many series are included. This is the high-power arm of the design.

**Scheduled narrative releases.** Central bank decisions and their accompanying statements, together with the associated press conferences where these occur. These are few in number but occupy taxonomy cells nothing else reaches, and several central banks release the statement and hold the press conference at a known offset, giving a within-day, within-participant contrast between a text release and an extemporaneous one.

**Scheduled multi-dimensional releases.** Corporate earnings for a panel of large, liquid names, restricted to those reporting during market hours so that the intraday adjustment is observable rather than compressed into an overnight gap.

**Unscheduled news.** Identified from a timestamped news analytics feed, filtered to high-relevance, high-novelty items on macro-sensitive topics. Arrival times come from the feed's first-mention timestamp. This arm is the noisiest and will be treated as exploratory.

**Placebo sample.** Matched clock windows on days with no scheduled release and no high-relevance news item.

A practical note on sample construction: the cross-announcement design means the unit of analysis is an *event*, and events are unbalanced across cells by construction: there are hundreds of inflation prints and a handful of press conferences. Specifications will therefore report both unweighted results and results weighting cells toward balance, and the primary contrasts will be tested within event day where the design permits.

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## 7. Data Requirements

**Market data.** The mechanism analysis (Q1, Q4) requires trade-and-quote data with sub-second exchange timestamps and, ideally, book depth beyond the touch. Futures on broad equity indices and government bonds are the preferred instruments: they are the most liquid, they trade through every release window including the pre-open macro slots, and their data is available per-day rather than only as continuous history, which matters for cost. Liquid ETFs on the same exposures serve as the cross-venue counterpart for Q3.

The speed and location analyses (Q2, Q3) degrade gracefully to one-second bars. The mechanism analysis does not; without trade-level data and aggressor flags, Q1 is not estimable.

**Announcement metadata.** Release calendar with scheduled timestamps, realized values, consensus forecasts, forecast dispersion, and prior-period revisions. Commercial economic calendars provide all of these; the dispersion field is the one most often missing from free sources and is essential to H3.

**News data.** A timestamped, relevance-scored news feed for the unscheduled arm. Note that headline archives without intraday timestamps (daily or weekly news corpora, for instance) cannot support this analysis regardless of how rich their content is, because the entire design rests on knowing the arrival second. This is worth stating plainly because it is a common and expensive mistake.

**Timestamp integrity.** Exchange-side timestamps only, never vendor receipt timestamps. Cross-instrument alignment must be validated against two or three known high-impact events before any analysis is run. This validation step is not optional and should be treated as a deliverable in its own right.

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## 8. Methodology

### 8.1 Standardizing the surprise

Announcements are not comparable until their surprises are on a common scale. For releases with a forecast survey, use the standardized surprise

$$S_i = \frac{A_i - F_i}{\sigma_i}$$

where  $A$  is the realized value,  $F$  the consensus, and  $\sigma$  the dispersion of forecasts for that release series. For policy decisions, where survey consensus is a poor summary of expectations, extract the surprise from short-horizon interest rate futures across the event window instead. For unscheduled news, no ex-ante expectation exists; these events are analyzed on magnitude and direction only and are excluded from specifications requiring a standardized surprise.

## 8.2 Decomposing the adjustment (Q1, primary)

For each event, partition the cumulative post-arrival log return into three parts:

1. **Pre-trade quote revision:** the midpoint change from the last quote before the arrival timestamp to the first quote after it, conditional on no intervening trade. This is the public-signal component.
2. **Order-flow component:** the cumulative fitted return from a regression of subsequent short-interval returns on contemporaneous signed order flow, estimated in event time. This is the strategic-learning component, and its slope coefficient is a Kyle-style price impact parameter measured within the event window.
3. **Residual:** everything else.

The mechanism share is component (1) divided by (1)+(2). This scalar is the project's central object. It is then regressed, across events, on the taxonomy variables:

*mechanism share ~ representation class + scheduled indicator + forecast dispersion + |standardized surprise| + instrument and time-of-day fixed effects*

Where a single event day contains two releases of different representation classes (a text release followed by a spoken one, for example), the within-day contrast is estimated directly with event fixed effects, which absorbs everything about that day's participants, volatility regime and market conditions.

## 8.3 Speed of discovery (Q2)

In event time from arrival, compute the variance ratio across sampling intervals; under an efficient random walk it equals one, and during active discovery it departs from one and converges as the price settles. The convergence horizon is the estimand. As a complement, estimate a state-space decomposition separating the efficient price from a transitory pricing error and track the decay of the pricing-error variance.

## 8.4 Location of discovery (Q3)

Estimate a vector error correction model on a cointegrated system of instruments tracking the same underlying exposure, at high sampling frequency over the first minutes after arrival, and

compute information shares. Report both the upper and lower bounds; the gap between them is typically wide at high frequency, and a single reported figure would overstate precision. A permanent-transitory component decomposition serves as a robustness check.

## 8.5 Liquidity dynamics (Q4)

Track depth at the touch, quoted and effective spreads, and the adverse-selection component of the effective spread in short buckets spanning the arrival. Compare the pre-arrival paths of scheduled and unscheduled events directly.

## 8.6 Asymmetry (Q5)

Interact the principal specifications with the sign of the standardized surprise, controlling for magnitude, and test coefficient equality.

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# 9. Deliverables

1. An event-time panel (event  $\times$  instrument  $\times$  time bucket) with full construction code and timestamp validation report.
  2. A harmonized announcement database with standardized surprises across release types, independently usable.
  3. A results table for the mechanism regression, which is the headline contribution.
  4. Supporting tables for speed, location, liquidity and asymmetry, each with placebo counterpart.
  5. A figure showing average cumulative response in event time, faceted by representation class, with placebo overlay.
  6. A written report.
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# 10. Timeline

## Weeks

## Milestone

- |   |                                                                         |
|---|-------------------------------------------------------------------------|
| 1 | Data procurement; timestamp validation on a pilot event set             |
| 2 | Announcement database and surprise standardization across release types |

- 3            Event-time panel on the pilot sample; every table and figure drafted end-to-end
- 4            Extension to the full sample
- 5–6        Mechanism decomposition and cross-announcement regression (the core contribution)
- 7            Speed and location analyses; placebos and robustness
- 8            Writing and figures

Eight weeks is tight enough that the schedule protects the primary question at the expense of the secondary ones. Q1 keeps two full weeks; the liquidity, asymmetry and subperiod analyses (Q4, Q5) are treated as extensions to be attempted only if weeks 5 to 7 run clean. Week 3 is the checkpoint that matters: if a complete draft of the results cannot be produced on the pilot sample by its end, the correct response is to narrow the announcement taxonomy rather than to compress the later weeks.

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## 11. Risks and Mitigations

**Data cost.** Purchasing tick data per event day rather than as continuous history reduces cost substantially. The project degrades gracefully: Q2, Q3 and Q5 survive on one-second bars, and only Q1 and Q4 strictly require trade-level data.

**Timestamp misalignment.** The single largest threat to validity, and the one most likely to produce a plausible but wrong result. Mitigated by exchange-side timestamps only and by explicit pre-analysis validation.

**Unbalanced taxonomy cells.** Narrative and extemporaneous releases are rare relative to numeric ones. Mitigated by leaning on within-day contrasts where they exist, reporting both weighted and unweighted specifications, and being explicit that the narrative arm is lower-powered.

**Unscheduled news identification.** Arrival timestamps for unscheduled news are noisier than for scheduled releases, and relevance filtering involves judgment. This arm is pre-designated exploratory; the primary results do not depend on it.

**Regime heterogeneity.** Market structure, volatility regimes and the relative importance of individual releases all shift over any multi-year sample. Results reported by subperiod as well as pooled.

**Specification search.** With many announcement types and many response measures, the space of testable relationships is large enough to guarantee spurious findings. Mitigated by fixing the hypothesis list in advance, reporting all pre-specified tests including nulls, and treating anything discovered outside that list as exploratory.

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## 12. Scope Discipline

The first pass runs one instrument and one announcement type end-to-end, from arrival timestamp through mechanism share, producing every table and figure in draft form before the sample widens. The cross-announcement design is what makes the project interesting, but it is also what makes it easy to spend six weeks building infrastructure and never reach a result. Build the narrow version first, then widen.

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## 13. Selected References

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